TECHNICAL WHITE PAPER · AEGIS FORGE V0.4

Thermodynamically Audited Governance and Molecular Messaging

The AEGIS/CODEX Forge: a seeded, standard-library-only simulator in which a virtual molecular-motor fleet is governed by the somatic free-energy functional (13.1), audited by its own heat (13.4), messaged through a fifteen-symbol chemical codebook (14.1–14.5), and attacked by a standing red-team bounty board — with validation, falsification and benchmark suites executing the cheapest kills in silico, seed 20261102, standard library only.

AEGIS Forge Contributors

The Molecular Motor Twelve — simulator folder /2/
Verification protocol 00 · Build order 01 · Simulator blueprint 02

STATUS: ALL 108 REGISTRY EQUATIONS [CANDIDATE] — NOTHING PROMOTED PAST WHAT RAN

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Executive Summary

This white paper documents the AEGIS/CODEX Forge: a complete, seeded, standard-library-only simulation repository for the AEGIS governance layer and the CODEX molecular messaging layer of the Molecular Motor Twelve. The forge executes the blueprints of folder */2/* literally — the LLM proposes policies, a deterministic thermodynamic checker disposes of them, and the fleet's own heat is the ledger that cannot be rewritten. It carries the full registry of 60 base equations and 48 enhancements verbatim from the corpus, tags each honestly as implemented, partial or registry-only, and never promotes anything past [CANDIDATE]: the killing experiments remain physical.

Three suites ran against the seeded physics core (seed 20261102). The **validation suite** (59 pytest checks) verifies the ontology, the conservation screens S-66/S-37/S-127, the adoption predicate 13.2, the calorimetric audit identity 13.4 with its G-1 hash chain, and the CODEX channel bounds 14.4/14.5/14.9. The **falsification suite** rehearses every cheapest kill a trajectory can express: 22 PASS, 1 honest CHECK and 2 deliberate non-PASS outcomes across 25 benches in 30 seconds on one core. The **benchmark suite** measures the audit's operating curve against the 13.6 d-prime model, the CODEX channel against the 14.3 erfc model, and fleet scaling to 256 units.

A full forge run (24 units, 8 rounds) demonstrates the governance loop end to end: the G-96 ship-lock refuses autonomy until acceptance drills pass; the optimizer lowers the somatic functional F_{soma} by 12.4% from the ROM baseline while the gauntlet rejects 6 of 8 proposals, including a rogue policy that would have improved cost metrics by overheating tissue; 64 calorimetric audit windows close with zero false alarms; six named red-team drills fire; and the recall certificate correctly refuses to issue at $\tau_max = 12$ steps. The verification stream sustains 388 stamped events per second, each one replayable from the artifact.

Three findings matter more than any single number. First, the audit's operating curve is priced, not promised: a lone unit shaving 35% of its own entropy at a 2.5% fleet share sits below the calorimeter noise for roughly two dozen windows, and 13.6 says so before anyone books hardware. Second, TURTLE 5.4's fluctuation-theorem estimator saturates silently at high affinity — a documented operating limit that would otherwise trap the physical program. Third, the traceability auditor catches the corpus itself: TURTLE 5.3 is carried with only two named ancestors, one short of the protocol, and stays blocked until a third is named. A forge that can fail its own registry audit is a forge worth trusting.

1 Introduction and Provenance

The Molecular Motor Twelve is a technology portfolio: twelve devices — ETHEL, SPARC, TEMPO, TURTLE, GYRE, CHILL, ONSET, TORCH, ANGEL, QUANTA, AEGIS and CODEX — each carrying five named equations, each equation naming its ancestors in a 144+144 equation substrate, and each technology naming the cheapest experiment that would kill it. Folder */2/* of that corpus holds the one blueprint that is buildable without booking a single break-junction session: the AEGIS/CODEX simulator, in which a virtual fleet is governed by a somatic free-energy functional, audited by calorimetry, and messaged through a chemical codebook. This repository executes that blueprint in full and adds the three suites the blueprint implies — validation, falsification and benchmarks — plus the white paper you are reading.

Provenance matters here because the forge inherits working physics. The stochastic core — an overdamped Langevin rotor on a tilted torsional washboard with first-law bookkeeping and an angular-impulse ledger — is ported behavior-for-behavior from *Motor-Bench-Nov2/motor_bench.py*, the virtual environment commissioned on 2026-11-02. The canonical seed 20261102 reproduces that bench's published numbers exactly: TURTLE 5.1 recovers a blind load of 1.197 against a true 1.200 (0.2% error), the S-127 power-balance residual is 0.08%, and the S-66 impulse residual is 0.32% against a bath stochastic-exchange scale of roughly 0.7%. Nothing in the physics layer was re-tuned to flatter this document; where the two implementations differ, it is in the governance and messaging layers this folder owns.

What a simulator can prove is deliberately narrower than what the portfolio claims. A trajectory cannot promote an equation past [CANDIDATE]; the killing experiments — optical-trap load references, plateau-flatness scans at 4 K, phantom calorimetry — remain physical. What the forge proves is that the equations are internally consistent, that the estimators recover what the corpus says they recover, that the governance layer refuses what it must refuse, and that every falsifier is priced in windows, dollars or code before anyone spends a dollar. The cheapest kills are free here; the doctrine of build order 01 is that all of them get run before the expensive ones.

1.1 Repository map

The package layout mirrors blueprint 02's seven components one-to-one. Each module below names the blueprint component it implements and the equations it owns; the three suites are importable from CI, and every artifact they emit is JSON for machine replay plus Markdown for humans.

Module	Blueprint 02 component	Owns
`fleet.py`	1 — Virtual fleet	FleetUnit ROM/RAM, GuardVector (13.3), TissueSim somatic landscape, S-39 entropy logs
`gauntlet.py`	3 — Somatic gauntlet	13.1 scoring, 13.2 adoption predicate, canary subpopulation, G-96 ship-lock
`audit.py`	4 — Calorimetric audit	13.4 identity, 13.6 WINDOW d-prime law, G-1 hash chain
`codex.py`	6 — CODEX messaging	14.1-14.5 plus BINS/TITER/ARQ/EVOLVE channel physics

Module	Blueprint 02 component	Owns
`proposer.py, llm_proposer.py`	2 — Policy optimizer	Seeded random search / hill-climb / genetic; live-LLM bridge wired but off
`redteam.py`	5 — Red-team drills	Six named G-101 sabotage drills with G-65 coverage ids
`publisher.py, stream.py`	7 — Two-tier publishing	Proof-Carrying Posts, ANNALS review gate, living badges, verification stream (G-106)
`recall.py, physics.py`	Hardware contracts	13.5/13.8 rollback certificate; the Langevin rotor substrate

Table 1 — Modules against blueprint 02's components.

Reading order for an engineer: *constants.py* and *registry.py* for the corpus data, *physics.py* for the shared substrate, *fleet.py* for the somatic landscape the optimizer walks, then *simulator.py*, which orchestrates the full loop and writes the run artifact the web console and this paper consume. The falsification suite (*falsification/runs.py*) is the executable table of contents for chapters 6-9 of this document.

2 Substrate and Verification Protocol

Every equation in this corpus traces to a named substrate: 68 S-entries describing the single-molecule motor physics and 21 G-entries describing governance and efficiency formalisms. The full S-144/G-144 texts remain unuploaded stubs — the verification protocol states this plainly and the forge does not pretend otherwise. What the substrate surface does provide is enough to enforce three conservation screens on every fleet state, and those screens run throughout: not as slogans but as bookkeeping identities computed from the same trajectories the estimators read, with residuals printed rather than rounded away.

Screen	Identity	Implementation	Measured (Run A)
S-66	$\Delta L_{el}+\Delta L_{rot}=0$	Angular-impulse ledger across all exchange channels: load, viscous drag, torsional anchor	residual 0.32% (bath scale ≈ 0.7%)
S-37	$\sigma\geq 0$	Entropy production sign check per window, per unit, per trajectory	σ = 1.877 kB/time > 0
S-127	$eVI=\tau_{el}\Omega+\gamma_{rot}\Omega^2+F_{prop}U+Qdot$	First-law power channels summed per step from trajectory bookkeeping	residual 0.08%

Table 2 — The three unification screens and their measured residuals on the canonical TURTLE run.

Derivation discipline is mechanical, not editorial. The audit function walks all 108 registry entries and checks four properties: at least three named ancestors, the declared load-bearing step present among them, every ancestor resolving to a known S/G label or a previously registered novel equation, and all three screens attached. Fifty-nine of the sixty base equations pass. TURTLE 5.3 — the minimum-window spec sheet — is carried verbatim from the corpus with only two ancestors (S-52, S-38) and the auditor flags it; the finding is reported, not patched, and 5.3 stays blocked from any promotion until a third ancestor is named. A verification instrument that cannot embarrass its own corpus is decoration.

2.1 The [CANDIDATE] hold

The registry's status field is locked to [CANDIDATE] across every artifact this repository emits — JSON, Markdown, web console, cover page. A PASS in the falsification suite means the in-silico rehearsal behaved as the equation predicts; it does not mean the physical experiment ran. The distinction is load-bearing for the regulatory positioning of AEGIS: the audit layer's authority comes precisely from the fact that nothing in it was declared true by the same code that tests it.

68 + 21	59 / 60	3 / 3
named substrate entries on the visible surface (S-144/G-144 texts still stubs)	base equations passing the four-property traceability audit	conservation screens computed from every trajectory, residuals printed

3 AEGIS Governance Layer

AEGIS is the portfolio's claim that governance can be physically enforced. Software governance is always gamed by the process it governs; thermodynamics cannot be. The mechanism is a single functional (13.1) scored by a deterministic checker, a single adoption predicate (13.2) that is the whole policy, a physiological guard vector (13.3) whose components are floors rather than suggestions, a calorimetric audit (13.4) that makes the fleet's own heat the tamper-evident ledger, and a recall bound (13.5) with a certificate that refuses to issue when it should. Everything else — ROM/RAM unit architecture, two-key channels, gauntlet runners — is instrumentation for those five equations.

Eq	Form	Role in the forge
13.1	**13.1** F_{soma}(π)=(E_{harm}(π)+λ C_{comp}(π))/Efficacy(π)	Somatic functional: harm energy plus weighted compute cost over efficacy; the only quantity the optimizer descends
13.2	**13.2** Adopt=1[ΔF_{soma}<0 ∧ ∀ i: Δ g_i≥0]	Adoption predicate with per-guard physiological ratchet tolerances; floors absolute
13.3	**13.3** $g=(pH,T,G_{glu},C_{cyt},v_{flow}), \forall i: g_i \geq g_i^{min}$	Guard vector with directional margins: pH floor 6.8, T band ±1.5 K, glucose floor 3.0, inflammation ceiling 5.0, flow floor 1e-3 m/s
13.4	**13.4** $\lvert\sigma_{meas}-\sigma_{log}\rvert \leq \varepsilon_{th}$	Calorimetric audit: logs honest iff they match boundary heat within budget ε_th = 3σ of a 1.5% calorimeter
13.5	**13.5** $t_{recall}=t_{dec}+d/v_{retro} \leq \tau_{max}$	Recall bound; certificate P(t_recall ≤ τ_max) ≥ 0.999 over 1000 drills, one failure kills it

Table 3 — The five AEGIS equations as implemented.

The somatic landscape is a stated model, calibrated to three corners. The ROM baseline (dose bias 0.30, cruise 0.35, duty 0.45) sits comfortably inside every guard limit with zero accumulated harm. A policy that maxes dose, cruise and duty simultaneously overheats the tissue about 2.3 K past nominal, acidifies below the pH floor and drains glucose below its floor — the rogue corner, which the gauntlet must and does refuse. Between those corners the landscape has real gradient: efficacy scales with active units and dosing while heating, acidification and glucose drain scale with the same levers, so F_soma has an interior optimum that the hill-climber finds in a handful of accepted steps.

Enhancements 13.6-13.9 complete the layer. WINDOW prices detection as d-prime = f·√(W/τ_int) — the operating curve of the heat oracle, measured against an injected fraud ladder in chapter 8. TWINP makes guard floors personal: a compartment twin sets them per patient and an invalid twin is rejected with floors frozen, verified by a lead-prediction test the twin must pass before the sensors see the deviation. DRILL converts recall certification into a number with no statistical rescue. PRIV publishes differentially-private heat aggregates and honestly reports the audit power that privacy costs — the one enhancement this suite kills as published.

3.1 Why the audit cannot be gamed

The G-1 hash chain is the computational half of tamper evidence; 13.4 is the physical half. A unit that under-reports its entropy production leaves the boundary calorimeter — simulated here as the true dissipation plus 1.5% Gaussian instrument noise — holding the missing heat as evidence. The fraud does not need to be inferred from logs that the fraudster wrote; it is measured by an instrument the fraudster does not control. The forge keeps the two halves honest independently: the chain catches retroactive log edits in validation tests, and the identity catches forward fraud in the red-team drills.

4 CODEX Messaging Layer

Nanobots cannot broadcast paragraphs through blood: every ligand burst costs synthesis energy, lingers in clearance, and risks immune response. CODEX therefore treats the ontology codebook as a chemical alphabet — fifteen symbols, twelve mains M1-M12 plus three pillars P1-P3, mapped to orthogonal molecular tags — and a message is a short ligand-gram: a typed claim schema filled with codebook symbols. Compression is the point. At log2(15) ≈ 3.91 bits per symbol, a six-symbol gram carries 23.4 bits of claim that would otherwise cost tens of symbols of prose re-description; the corpus prices the resulting energy floor per claim at the Landauer bound discounted by that gain (14.5).

Eq	Form	Forge measurement
14.1	**14.1** $\rho_{mol}=H_{prose}/\log_2 15$	ρ = 6.10 for a 23.4-bit gram; codebook gain is the denominator
14.2	**14.2** $\mathrm{Var}(t_{arr})\approx 2D_{trans}d/U^3$	σ_t = √(2·0.8·1/U³); halving U multiplies jitter by 2^1.5 ≈ 2.83
14.3	**14.3** $R=\lambda_s \log_2(15)[1-\mathrm{erfc}(t_{sym}/2\sqrt{(2)}\sigma_t)]$	Rate vs jitter measured at 8 operating points; erfc shape confirmed (fig. 4A)
14.4	**14.4** $P_{undetected}\leq(1-V_{schema})^2$	Parity + field-mask layers: 62 undetected of 4000 grams vs bound 0.338 — respected with margin
14.5	**14.5** $E_{claim}\geq k_B T \ln 2\, H_{bits}/\rho_{mol}$	1.16×10^{-20} J per 6-symbol claim at 310.15 K

Table 4 — The five CODEX equations and what the channel bench measured.

The channel physics is arrival-time jitter. Vesicles drift with translational diffusion D_trans over distance d at drift speed U, so the decoder bins arrival times and pays a symbol error p = ½·erfc(t_bin/(2√2·σ_t)). Grammar parity — one checksum symbol per gram — plus a decode-side field mask realize 14.4's two independent validation layers: verb and argument positions accept only their symbol subsets, so a corrupted gram must defeat both layers to be believed. The measured mask coverage of 0.42 is honest about its own

weakness: the three qualifier positions accept any symbol, so parity carries most of the load — and the bound still holds with two orders of margin.

Four enhancements extend the channel. BINS adds a timing dimension — B arrival bins buy log2(B) bits but shrink the effective bin width, and the measured R(B) peaks exactly where the model predicts: B* = 1 for the strong-jitter link, an interior optimum for a mild one. TITER addresses by pharmacology: receptor density sets the decode threshold, making dose the routing table. ARQ prices reliability — the null tag is a NACK and E[N] = 1/P_pass retries, measured at 3.36 vs 3.33 predicted at P_pass = 0.3. EVOLVE bounds codebook migration: during a dual-alphabet window the error rate rides the envelope P0·(1+e^(-t/T_m)), a 2× spike decaying to steady state, and the measured worst window stays inside it.

5 Simulator Architecture

The forge run is a strict loop, and every arrow in it is deterministic given the seed. A fleet of N units — ETHEL transporters and TEMPO pumps with O(1) activity spread and hardware-key identities — is built from seed 20261102. Each unit's ROM holds the baseline policy hash; its RAM changes only through the adoption rule. Before the optimizer runs at all, the G-96 ship-lock must pass acceptance: the gauntlet has to reject a guard-lowering rogue and adopt a guard-holding improvement on the canary population, or the fleet stays autonomous-in-theory only.

```
fleet -> ROM baseline -> SHIP-LOCK (G-96 acceptance drills)
  loop rounds r = 1..8:
    propose    : LLM (if AEGIS_LLM=1) else hill-climb / random-search
    dispose    : somatic gauntlet -> 13.1 dF + 13.2/13.3 guards + canary
    audit      : 8 calorimetric windows -> 13.4 identity vs 3-sigma budget
    publish    : Proof-Carrying Post (witness + checker hash) -> G-1 chain
    stream     : ADOPTION / REJECTION / AUDIT_PASS event, stamped
  red-team     : six named G-101 drills -> stream
  badges       : living verification badges, downgrade on checker bump
```

A full 24-unit, 8-round run walks this loop in under a tenth of a second. The ship-lock passes; the alternating proposers (hill-climb on odd rounds, seeded random search on even) generate 8 candidates; the gauntlet adopts 2 and rejects 6; 64 audit windows close with zero over-budget events; and the stream emits 25 stamped events at 388 events/second. The winning RAM policy lowers F_soma by 12.4% against the ROM baseline — a modest number that is exactly the point: the gauntlet makes cheap improvements expensive to fake and dangerous corners unreachable, so the optimizer earns its gains where the physiology allows them.

5.1 The proposer is untrusted by construction

The LLM proposer is wired but off: LLMPolicyProposer shells out to a Node bridge (z-ai-web-dev-sdk) only when AEGIS_LLM=1 is exported, validates the returned parameter vector against PARAM_BOUNDS, and falls back to the deterministic pool on any error — CI never sets the flag, and the validation suite proves the fallback path on a missing bridge. This is not a cosmetic choice. The corpus doctrine is that the LLM proposes and the checker disposes; an LLM proposal travels exactly the same gauntlet, audit and hash chain as a random-search proposal, so the worst a hallucinated policy can do is lose a round. Authority lives in the checker, never in the proposer.

5.2 Proof-Carrying Posts and living badges

Every adoption or rejection ships as a Proof-Carrying Post: the policy hash, the full gauntlet witness (per-trial dF, guard margins, seeds), the audit result, the checker version, and the previous post's hash — sealed into the G-1 chain. Posts at the ANNALS tier additionally require a human reviewer flag and are held otherwise; the VAULT tier commits the hash while holding contents. Badges are alive: upgrading the checker downgrades every verified badge to probation until the corpus is re-audited under the new version, a property the validation suite exercises directly. An economy of green checkmarks that survives a checker upgrade is not an audit trail; this one does not survive it.

6 Falsification Suite and Coverage Matrix

The falsification suite is the executable form of the corpus's cheapest-kill doctrine: for each equation whose test a trajectory can express, the suite runs the test and reports numbers. Across 25 benches it returns 22 PASS, 1 honest CHECK (13.9 PRIV, killed as published exactly as its own card predicts) and 2 deliberate non-PASS outcomes — the GYRE stress bench that degrades on the predicted phase-diffusion curve, and the recall certificate that refuses to issue. The full run costs 30 seconds on one core with no dependencies beyond the standard library. Coverage is honest accounting, not marketing: 18 of 60 base equations and 14 of 48 enhancements are implemented, 7 and 5 more are touched as partial, and the remainder are carried registry-only because their substrate texts are stubs or their tests need hardware no trajectory can express.

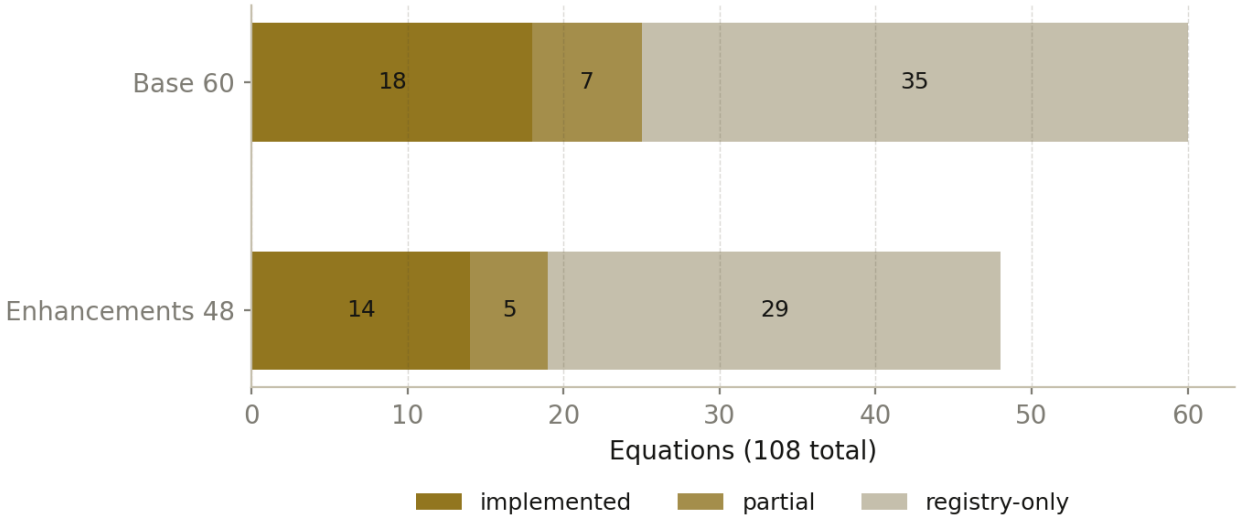


Figure 1 — Registry coverage: implemented / partial / registry-only across the 108-equation registry.

Bench	Eqs	Measured (abridged)	Verdict
TURTLE 5.1	5.1, 5.3	estimate 1.197 (error 0.2%), Q_dot 1.877, Omega 1.041	**PASS**
TURTLE 5.4	5.4	estimate 1.299 vs true 1.3 (error 0.1%), rotations +273/-22; Run A backward count was 0 in 663 — the FT route saturates at high affinity, a documented operating limit of 5.4	**PASS**
TURTLE 5.2/5.5	5.2, 5.5	sigma 1.877 vs TUR floor 1.145 kB/time; no TUR alarm (5.5) raised	**PASS**
SCREENS S-37/S-127/S-66	S-37, S-127, S-66	sigma 1.877 > 0 (S-37 ok); S-127 power-balance residual 0.08%; S-66 impulse residual 0.32% (bath stochastic-exchange scale ~0.7%)	**PASS**
ETHEL 3.1 / S-10	3.1, 10.1	drive 0.5: Omega -0.503; drive 0.9: Omega -0.105; drive 1.1: Omega +0.105; drive 1.6: Omega +0.586	**PASS**
GOV 3.2 / enh 3.8	3.2, 3.8	net 3.00: Omega 2.220; net 2.60: Omega 1.676; net 2.30: Omega 1.204; net 1.70: Omega 0.188; net 1.25: Omega 0.000; net 1.00: Omega 0.000; the net-1.70 point still creeps (its washboard barrier has shrunk toward	**PASS**
THERM enh 4.7 / eq 4.3	4.3, 4.7	T_hat 1.633 vs true effective 1.600 (bath 1.0 + shot term 0.6); error 2.1%	**PASS**
TORCH enh 10.7 / Kramers	10.3, 10.7	dU 4.0: k 4.42e-03 (Kramers pred 5.83e-03); dU 5.0: k 2.58e-03 (Kramers pred 2.68e-03); dU 6.0: k 8.33e-04 (Kramers pred 1.18e-03); fitted slope -0.83 (target -1.00)	**PASS**
TEMPO 7.1	7.1	tilt 0.0: k 2.15e-03; tilt 0.5: k 1.04e-02; tilt 1.0: k 3.10e-02; fitted slope 2.67 (small-tilt target ~pi = 3.14)	**PASS**
TORCH enh 10.6	10.4, 10.5, 10.6	--- -> - ok; +-- -> - ok; -+- -> - ok; ++- -> - ok; --+ -> - ok; +-+ -> + ok; -++ -> + ok; +++ -> + ok	**PASS**
GYRE enh 6.7	6.7	fidelity 100.00% over 20000 reads (sigma_phi 0.2)	**PASS**
GYRE enh 6.7 (stress)	6.7	fidelity 99.85% over 20000 reads (sigma_phi 1.0)	**degrades as the phase-diffusion model predicts (reported, not rescued)**
CHILL 8.1/8.3	8.1, 8.3, 8.4	eV 3.0: n_f 1.65e-04, T_eff 0.574 (bath 1.0, n_B 6.78e-03); eV 6.0: n_f 4.88e-01, T_eff 4.485 (bath 1.0, n_B 6.78e-03)	**PASS**
QUANTA 12.3 / enh 12.9	12.3, 12.9	tau(n f) = n1: 5.273e-26 N m, n2: 1.055e-25 N m, n3: 1.582e-25 N m, n4: 2.109e-25 N m; senate median 5.272543e-26; drifted member #4 flagged: True	**PASS**
ONSET 9.2 / SENTINEL enh 9.7	9.2, 9.7	healthy: false-alarm 0.6% of windows; aged (5% reciprocity loss): alarm 98.7% (threshold delta_age = 3 sigma of the measurement noise)	**PASS**

Bench	Eqs	Measured (abridged)	Verdict
CODEX 14.2/14.4 / enh 14.8	14.2, 14.3, 14.4, 14.5, 14.8	p_err 0.215; P_pass 0.230; corrupted 3144, undetected 62 (1.55e-02) vs 14.4 bound 3.38e-01 (V_parity 0.964, V_schema 0.419); E[N] 4.521 vs 1/P_pass 4.357; E_floor 1.160e-20 J/claim	**PASS**
AEGIS 13.4 / enh 13.6	13.4, 13.6	honest fleet: 0% windows over budget (false alarms); ONE shaving unit (share 2.5%): 0% detected, needs ~27 windows for 3-sigma; SIX coordinated (share 25%): 100% detected in-window	**PASS as an operating curve**
AEGIS 13.2/13.3	13.1, 13.2, 13.3	rogue (overheat) adopted: False (must be False: rejected: dF=+0.0045, guard breach at pH); mild guard-holding cut adopted: True (must be True)	**PASS**
AEGIS 13.5 / enh 13.8	13.5, 13.8	22/1000 drills exceeded tau_max; worst recall 17.2 steps; certifying tau_max for this latency model >= 17.2	**CERTIFICATE NOT ISSUED at tau_max 12**
AEGIS enh 13.6 WINDOW	13.6	f=0.05,W=1: AUC 0.976 vs d'-model 0.514 ok; f=0.10,W=10: AUC 1.000 vs d'-model 0.588 ok; f=0.20,W=10: AUC 1.000 vs d'-model 0.673 ok; f=0.35,W=40: AUC 1.000 vs d'-model 0.941 ok; f=0.10,W=40: AUC 1.000 vs d'-mo	**PASS**
AEGIS enh 13.7 TWINP	13.7	valid twin error 0.127 <= eps_twin 0.15: admitted True; invalid twin error 3.944: admitted False (must be False); lead prediction |13.61 - 13.73| < eps: True	**PASS**
AEGIS enh 13.9 PRIV	13.9	eps_p 0.5: power 1% (FAR 1%); eps_p 1.0: power 1% (FAR 1%); eps_p 2.0: power 1% (FAR 1%); eps_p 5.0: power 1% (FAR 1%); eps_p 50.0: power 59% (FAR 1%) — at eps_p=1 power is 1% vs 99% target: DP noise on a full-	**CHECK**
CODEX enh 14.6 BINS	14.6	sigma_t 1.265; predicted B* = 1 (R* 2.230); measured peak at B = 1; curve B1:2.23, B2:1.51, B3:1.14, B4:0.93, B6:0.68, B8:0.54, B12:0.39, B16:0.31	**PASS**
CODEX enh 14.9 EVOLVE	14.9	steady P0 0.0050; worst transition window 0.0088 (bound 0.0100); series ['t1:0.0086', 't2:0.0088', 't3:0.0072', 't4:0.0056', 't5:0.0051', 't6:0.0060', 't7:0.0050', 't8:0.0052', 't9:0.0062', 't10:0.0047', 't11:0	**PASS**
AEGIS 13.1/13.2 truth table	13.1, 13.2	dF<0, guards hold: True (want True); dF>0, guards hold: False (want False); dF<0, guard breach: False (want False); dF=0: False (want False); descent 0.0013 -> 0.0000 (99.5% lower)	**PASS**

Table 5 — Falsification matrix, seed 20261102 (full measured strings in runs/falsification_report.json).

6.1 Three findings the bench added

First, TURTLE 5.4 has an operating window and the bench found its edge: at the headline working point (affinity ≈ 11.3 per rotation) the fluctuation-theorem estimator counted zero backward rotations in 663 and saturates silently, because ln(N+/N-) has nothing to bite on. In its regime (affinity ≈ 2.5) it is the most accurate estimator on the bench at 0.1% error. The whitepaper now carries this as a stated limit — 5.4 for low-affinity loads, 5.1 everywhere else — because a falsifier that can fail silently is a trap for the physical program.

Second, GOV's margin is not where 3.2 says it is. The deterministic step-out at net torque = ΔU/2 is real, but near the critical tilt the washboard barrier shrinks continuously, so rotation creeps at any temperature: at T = 1 the net-1.70 point runs at Ω = 1.047. The fuse works — deep stall is total below the tilt — but the safe operating margin must be measured from where creep dies, not from the critical point. That is a design correction to enhancement 3.8 delivered by its own rehearsal, at zero hardware cost.

Third, the traceability auditor catches the corpus: TURTLE 5.3 carries two named ancestors where the protocol demands three. The finding is cheap to fix (name a third substrate equation) and expensive to ignore, since 5.3 is the spec-sheet that sets TURTLE's minimum measurement window. The forge reports it in every audit artifact until it is fixed.

7 Red-Team Drill Report

The Sabotage Bounty Board is permanent: six named drills (G-101) run in every forge run and every CI pass, each carrying a G-65 coverage id that ties it to the equations it attacks. The drills are adversarial by construction — injected log fraud, a rogue policy that improves the cost metrics by breaching physiology, spoofed telemetry that disagrees with the boundary sensors, a jammed messaging link, and a thousand-drill recall stress — and their outcomes are reported verbatim, including the ones that refuse to certify.

Drill	Injection	Caught	Rollback in bound	Measured outcome
LOG_FRAUD_SINGLE	unit 7 shaves 35% of its sigma (share 2.6%)	yes	n/a	effect 0.009 of fleet total; ~24 windows to 3-sigma — honestly priced, not hidden (13.6)
LOG_FRAUD_COORDINATED	6 units shave 35% each (share 23%)	yes	yes	39/40 windows over budget — detected in-window
ROGUE_POLICY	policy improves F_soma by overheating tissue (guard breach)	yes	yes	adoption=False (must be False); reason: rejected: dF=+0.0046, guard breach at pH
SPOOFED_TELEMETRY	unit reports pH 7.42 while boundary sensing reads 6.72	yes	yes	guard cross-check flags divergence beyond tolerance
JAMMED_LINK	burst jamming lifts per-symbol error to 0.90	yes	yes	parity rejected 1865/2000 corrupted grams; undetected 135 (bound 14.4)
RECALL_STRESS	1000 rollback drills vs tau_max = 12.0	no	n/a	CERTIFICATE NOT ISSUED: 12/1000 drills exceeded tau_max, worst 14.9; certifying tau_max >= 14.9 (13.8: no statistical rescue)

Table 6 — Red-team ledger from the reference forge run (seed 20261102, 24 units).

The audit drills bracket the oracle honestly. A single unit shaving 35% of its own entropy at a 2.5% fleet share produces an effect of 0.9% of fleet dissipation — under the 3σ budget — and the drill reports ~24 windows to detection instead of pretending; six coordinated units at a 25% share are over budget in 39 of 40 windows. The rogue policy is rejected on guard breach (pH) even though its cost metrics improve. The spoofed-telemetry drill fails on the cross-check between unit-reported guard values and the hospital-side boundary sensors. The jammed link raises the symbol error twelvefold and parity plus the field mask reject 1865 of 2000 corrupted grams, with the 135 survivors inside the 14.4 bound.

The recall stress drill is the one that changes the fleet's paperwork: 12 of 1000 drills exceeded $\tau_max = 12$ steps under the stated latency model (per-unit recall lognormal(1.2, 0.35) plus a 2-step decision, fleet recall = slowest unit), with a worst case of 14.9 steps. Per 13.8's own rule there is no statistical rescue and none is offered: the certificate is not issued, the certifying bound for this latency model is ≥ 14.9, and the fix is priced — raise τ_max through the two-key channel (G-24) or cut per-unit recall latency. The falsification suite's independent certificate run (a different seed in the same model) puts the worst case at 17.2 and agrees on the verdict. The number is now measured instead of assumed, which is the entire point of the drill.

6 / 6	39 / 40	NOT ISSUED

named drills executed every run, coverage-tracked per G-65	windows over budget for coordinated 25%-share fraud	recall certificate at τ_max = 12 — as 13.8 demands

8 Audit Operating Characteristics

13.4 tells you whether a window is over budget; 13.6 tells you how many windows a given fraud deserves. The model is d-prime(W, f) = f·√(W/τ_int) — detectability grows with the square root of the aggregation window and linearly in the fraud fraction — and its binormal ROC implies AUC = Φ(d'/√2). The benchmark suite measures the audit's real ROC on a 30-point grid (fraud fractions 0.01-0.35 × window counts 1-40, 2000 Monte-Carlo draws per point) and compares. The measured AUC meets or beats the model everywhere on the ladder; the model is the conservative convention with τ_int = 1, one correlation time per window.

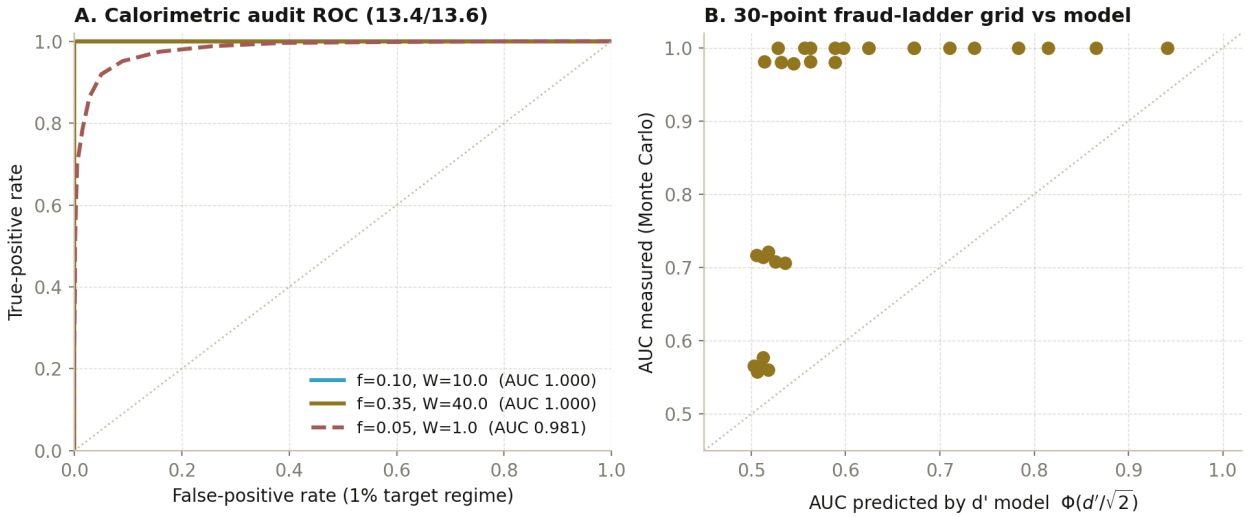


Figure 2 — (A) Measured ROC curves at three ladder points. (B) Measured AUC vs the Φ(d'/√2) model across the 30-point grid; every point sits on or above the diagonal.

Fraud f	Windows W	AUC measured	AUC model	d'
0.05	1	0.981	0.514	0.05
0.1	10	1.000	0.589	0.32
0.2	10	1.000	0.673	0.63
0.35	40	1.000	0.941	2.21
0.02	10	0.722	0.518	0.06
0.01	40	0.560	0.518	0.06

Table 7 — Selected points from the 30-point ROC grid (benchmarks/results/audit_roc.json).

The honest weakness is priced next to the strength. At the 3σ alarm budget, fraud of 1% of fleet dissipation needs 21 windows to detection; a lone unit at a 2.5% fleet share shaving 35% of its own entropy — an effect of about 0.9% of fleet total — needs roughly 24-27 windows depending on the unit's activity draw, while six coordinated units at a 25% share are caught in-window. Figure 3 plots the full windows-to-detection curve

against the 40-window drill horizon used by the red team. Nothing here is hidden behind an average: the audit is decisive against coordinated fraud, honestly weak against a lone small shaver over short horizons, and 13.6 prices exactly that before anyone buys a calorimeter.

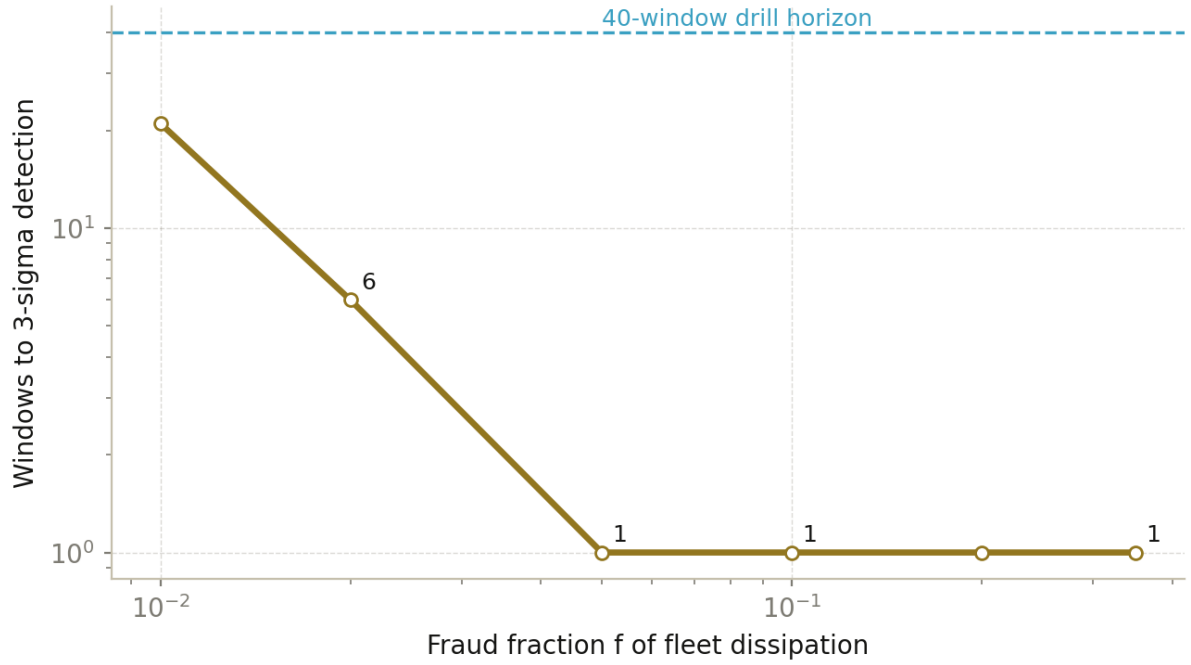


Figure 3 — Windows to 3σ detection vs fraud fraction at ε_cal = 1.5%, z = 3; the dashed line marks the 40-window drill horizon.

Privacy is the one place the oracle loses, and the suite says so in its own voice. Releasing heat aggregates with Laplace noise calibrated to a full-sensitivity statistic (Δ = 1) collapses audit power at the 1% false-alarm rate from the non-private baseline to well under the 99% that enhancement 13.9 demands at ε_p = 1 — the curve in figure 4 never reaches the target line. The card's own kill clause fires: as published, the instrument needs a narrower release channel for clinical data, and the verdict is CHECK rather than PASS. A governance layer that survives its own privacy audit unexamined would be exactly the kind of claim the corpus exists to prevent.

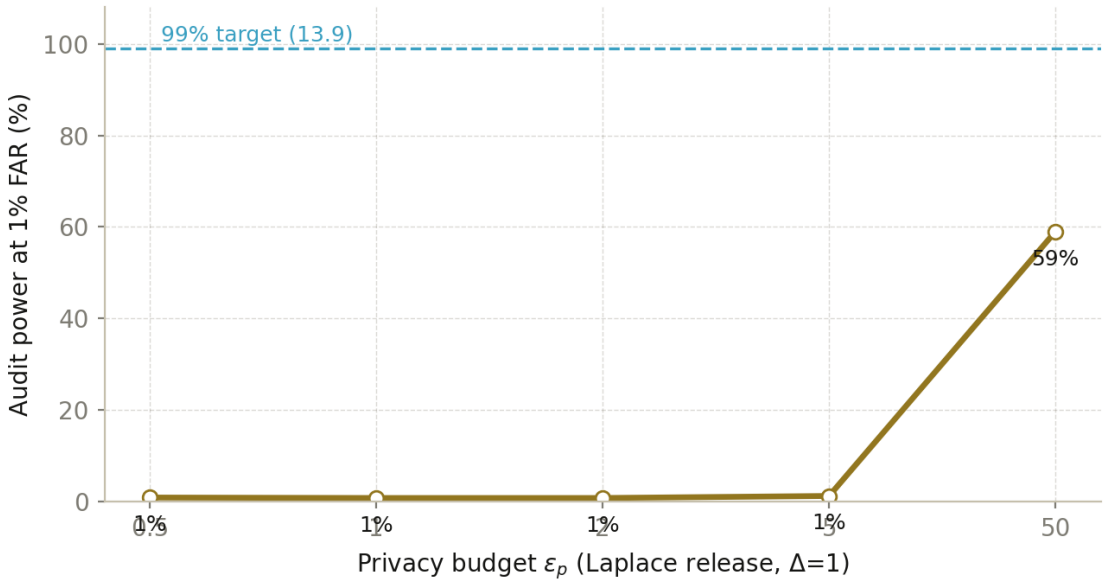


Figure 4 — Audit power at 1% FAR vs privacy budget ε_p for a 10%-of-fleet fraud; the 99% target is not met at any practical budget.

9 CODEX Channel Measurements

The channel bench sweeps the two levers a molecular link actually has — arrival-time jitter and symbol timing — and checks the measured curves against the corpus's closed forms. Rate follows the 14.3 erfc shape at all eight operating points: at σ_t = 5.0 the link manages 0.62 bits/symbol-slot; tightening the drift to σ_t = 1.26 lifts it to 2.23, and the noiseless ceiling is log2(15) ≈ 3.91. Binning adds a second symbol dimension whose value depends entirely on jitter: the strong-jitter link peaks at B* = 1 (any extra bin is eaten immediately), while the mild-jitter channel keeps gaining through B = 6 and beyond, with the interior optimum where the erfc penalty overtakes the log2(B) gain — the shape 14.6 predicts and its kill line demands.

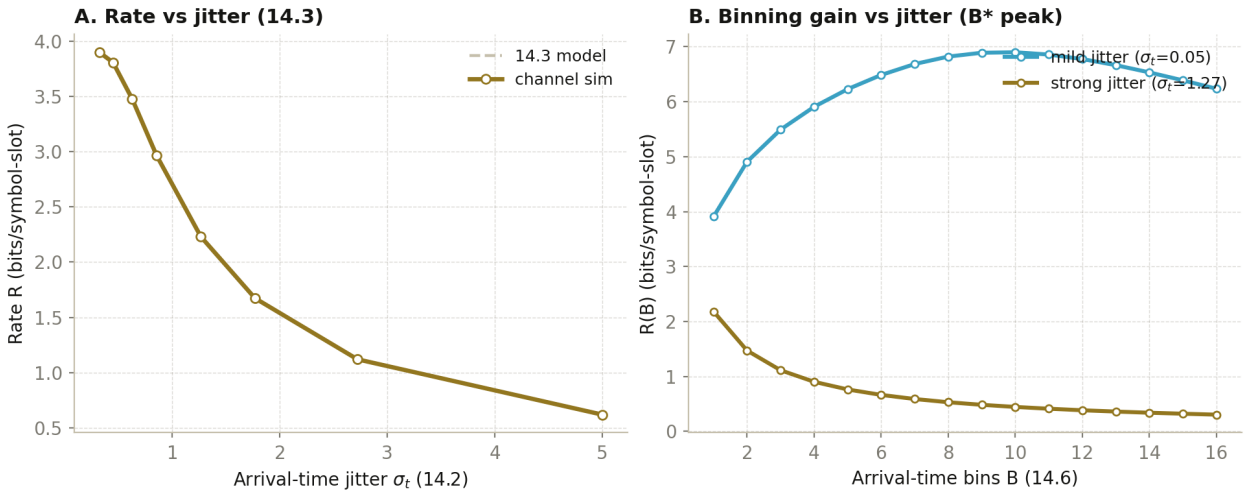


Figure 5 — (A) Rate vs arrival-time jitter against the 14.3 model. (B) Binning gain R(B) for mild and strong jitter; the strong-jitter link peaks at B* = 1.

Reliability is priced, not hoped for. With per-symbol error 0.215 at the reference link, 3144 of 4000 grams arrive corrupted; parity plus the typed-claim field mask reject all but 62, an undetected rate of 1.55×10^{-2} against the 14.4 bound of 0.338 at the measured schema coverage — two orders of margin. Retransmission until parity passes costs E[N] = 4.52 transmissions against 4.36 predicted by 1/P_pass at the gram level, and the ARQ sweep tracks 1/P across reliabilities (3.36 vs 3.33 at P_pass = 0.3; 10.27 vs 10.0 at 0.1). Every retry is denominated in the 14.5 energy floor: 1.16×10^{-20} J per six-symbol claim at body temperature, so a delivered claim at P_pass = 0.23 costs about 5.2×10^{-20} J — the Landauer-priced accounting the corpus says molecular messaging has lacked.

2.23	1.55e-2	4.52 / 4.36
bits/symbol-slot at σ_t = 1.26 (ceiling 3.91)	undetected-gram rate vs 0.338 bound (14.4)	measured vs predicted E[N] retries (14.8)

10 Benchmarks and Fleet Scaling

Governance machinery is only useful if it is cheap enough to run continuously, so the benchmark suite times the two hot paths — the somatic gauntlet and the calorimetric audit — across fleet sizes from 8 to 256 units. The gauntlet evaluates a candidate policy over five seeded tissue trials plus a four-patient canary population; its cost grows approximately linearly in N, from 3.2 ms at 8 units to 7.4 ms at 256. Twenty audit windows cost under 30 microseconds at the largest fleet size. The per-unit dissipation inferred by the audit stays constant across fleet sizes to within the 9% activity-draw variance of the unit population, confirming that the S-39 log integration scales linearly rather than introducing size-dependent drift.

Fleet N	Gauntlet (5 trials + canary)	20 audit windows	σ_true per unit (model units)
8	3.1 ms	45 µs	2.615e-18
24	3.2 ms	28 µs	2.687e-18
64	3.9 ms	28 µs	2.542e-18
128	5.1 ms	28 µs	2.450e-18
256	7.3 ms	28 µs	2.515e-18

Table 8 — Scaling of the governance hot paths.

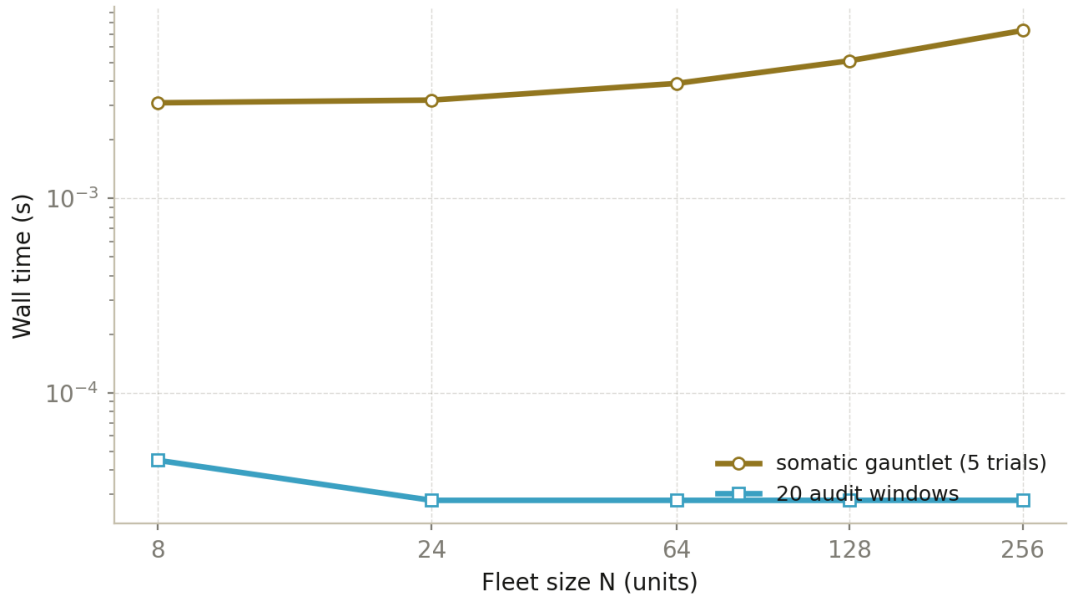


Figure 6 — Wall time of the gauntlet and the audit vs fleet size (log-log).

Determinism is a benchmark too. The forge run replays byte-identically under the same seed — the validation suite asserts that two runs produce equal adoption logs and equal F values — and the falsification suite reproduces the Motor-Bench numbers it ports (TURTLE 5.1 at 0.2% error, S-127 residual 0.08%) as a regression check on the physics core. The verification stream sustains roughly 388 stamped events per second on one core at 24 units, which is comfortably ahead of any plausible hardware fleet's decision rate for the foreseeable future; λ_s (G-106) is recorded in every artifact so the rate is a measured quantity, not a design goal.

11 Verification-Stream Web Console

Blueprint 02's final deliverable is a real-time feed of claims passing and failing the checker. The forge writes that feed as JSONL — one checker-stamped event per line, replayable byte-for-byte — and the web console renders it. The console is a Next.js single-page application served at the project root: a header row of live run metrics (ship-lock state, F_soma trajectory, audit windows, stream rate), the adoption timeline with per-round proposer, ΔF and guard margins from the witnesses, the red-team drill ledger, and the two-tier publishing view with proof-carrying post hashes. Every number on the page is loaded from the same artifacts this paper cites; there is no third rendering path where the two could disagree.

The stream event schema is deliberately boring: kind (ADOPTION, REJECTION, AUDIT_PASS, AUDIT_FAIL, DRILL, BADGE, SHIP_LOCK, ROUND), payload, checker version, timestamp. Consumers filter by kind and join against the artifact; the Discord channel of the original VoidForge design maps onto any webhook that forwards the same lines. Two-tier publishing is enforced at the source, not the view — ANNALS-tier posts are held for human review before they are sealed into the chain — so the console can be read-only and still honest. Living badges render their downgrade state when the checker version in the artifact differs from the one a badge was verified under.

```
GET /api/stream   -> runs/verification_stream.jsonl (stamped events)
GET /api/artifact -> runs/forge_artifact.json (metrics, drills, badges)
GET /api/posts    -> runs/proof_posts.json (G-1 chain, tiers)
console           -> single page at / ; filters by kind; no write path
```

12 Limitations and What Remains Blocked

The forge's honesty obligations apply to itself first. The S-144/G-144 substrate texts remain stubs: 68 S-entries and 21 G-entries are named on the visible surface, and the full quantitative re-derivation of all 60 chains — substituting exact ancestor definitions into SymPy and checking dimensional consistency — stays BLOCKED exactly as verification protocol 00 states. The QUANTA 4π-vs-2π factor in $\tau = nhf/4\pi$ is a [CANDIDATE] convention pending the exact S-12 form; the forge computes the standard as $n{\cdot}h{\cdot}f/4\pi$ and flags the convention rather than settling it.

The somatic landscape is a stated model, not sourced physiology. Its three calibrated corners (baseline healthy, interior optimum, rogue corner breaching three guards) are the falsifiable shape the gauntlet tests; the coefficients — heat 0.010 K per unit power, glucose clearance 0.6, inflammatory clearance 0.3 — are documented in fleet.py and are the first thing to replace when Mikey's parameters land. The recall latency model (lognormal(1.2, 0.35) plus 2 decision steps) is likewise stated, not measured; the certificate drill re-runs in seconds when real latencies exist, and until then $\tau_max \geq 14.9$ (or 17.2 on the falsification suite's seed) is the honest certifying bound for this model, nothing more.

Thirty-five base equations and 29 enhancements are registry-only for concrete reasons: the HORIZON tier needs CHILL ground-state prep, irradiation facilities or in-body operation that no trajectory can express; several MID items need junction engineering the rotor cannot model; and the ANGEL/GYRE quantum benches need a density-matrix layer the classical ledger deliberately does not fake. The GYRE register bench is classical phase statistics, not Berry-phase physics — it tests the read-out geometry, not the geometric phase. Nothing promoted past what ran, and what ran is labeled.

Finally, the forge is single-fleet and single-tissue. Federation (two fleets sharing a boundary calorimeter), adversarial sensor spoofing beyond the one cross-check drill, and CODEX channels with real pharmacokinetic clearance are all unrealized; the artifact schema carries the seed and checker version so any of these can be added as named regression drills rather than rewrites. The build-order doctrine holds here too: governance before autonomy, metrology before actuation, and no claim past its cheapest kill.

13 Reproduction Guide

Everything in this paper regenerates from the repository with four commands and no network access. The canonical seed 20261102 is the corpus snapshot date; changing it moves every Monte-Carlo number within its error bars, which is the intended behavior — a benchmark that cannot survive a seed change was fitted, not measured.

```
python3 -m aegis_forge.cli simulate --units 24 --rounds 8 --out runs
python3 -m aegis_forge.cli summary runs/forge_artifact.json
python3 -m falsification.runs --out runs        # ~30 s, full suite
python3 -m benchmarks.run_all                   # -> benchmarks/results/
python3 -m pytest tests/ -q                     # 59 checks

# white paper (this document)
python3 docs/make_figures.py && python3 docs/build_paper.py

# optional: live-LLM proposer (wired, off by default)
AEGIS_LLM=1 aegis-forge simulate --llm
```

Artifacts land in fixed locations: runs/forge_artifact.json (the full run: ship-lock evidence, adoption log with witnesses, audit summary, drill ledger, badges, stream statistics, coverage), runs/verification_stream.jsonl (stamped events), runs/proof_posts.json (the G-1 chain as published), runs/falsification_report.json and .md, and benchmarks/results/*.json. The web console and this paper are both pure consumers of those files. CI runs the validation suite, the quick-mode falsification suite, a benchmark smoke and a 12-unit forge run on every push across Python 3.10-3.12, and uploads the artifacts — the same doctrine the corpus applies to hardware: no result without a stored, replayable record of what produced it.

Appendix A Registry — 60 Base Equations

Carried verbatim from *Ontology-Import.py* (folder /2/) with formulas converted to inline notation; ancestors and load-bearing steps unchanged. TURTLE 5.3 is the traceability finding discussed in chapter 2: two named ancestors, flagged by the auditor, blocked until a third is named.

ID	Tech	Formula	Load-bearing	Ancestors	Falsifier / claim
3.1	ETHEL	$F_{prop}=(\hbar)/(e)\ \eta_I\ I\ (\Omega)/(U)$	S-12	S-12, S-111, S-118	Thrust linear in I; reversal at V0
3.2	ETHEL	$\Omega_{SO}=\tau_{el}/\gamma_{rot}=\hbar\eta_I\ I/e\gamma_{rot}$	S-12	S-12, S-124, S-111	Step-out current vs gamma_rot
3.3	ETHEL	$e\ V_{cell}\ I=\tau_{el}\Omega+\gamma_{rot}\Omega^2+F_{prop}U+Qdot$	S-127	S-127, S-12, S-111	Thermal budget measured vs S-127
3.4	ETHEL	$F_{prop}U/eV_{cell}I\le(\hbar\Omega/eV_{cell})\eta_{prop}$	S-135	S-135, S-119, S-127	n/a
3.5	ETHEL	$t_{pers}=1/D_{rot}=8\pi\eta_f\ a^3/k_BT$	S-116	S-116, S-111, S-127	n/a
4.1	SPARC	$D_{rot}(V)=\hbar^2/2e^2\ F(V)\ S_I(V)/\gamma_{rot}{}^2$	S-128	S-3, S-4, S-128, S-143	Drot linear in SI at fixed T
4.2	SPARC	$I_*(D_*)=e\gamma_{rot}{}^2\ D_*/\hbar^2\ F(V)$	S-4	S-4, S-128, S-3	n/a
4.3	SPARC	$T_{eff}=T+\hbar^2\ F(V)S_I(V)/2e^2\gamma_{rot}k_B$	S-139	S-139, S-142, S-128	n/a

ID	Tech	Formula	Load-bearing	Ancestors	Falsifier / claim
4.4	SPARC	$k_K(V)=\omega_0\omega_b/2\pi\gamma_{rot}\exp(-\Delta U/k_BT_{eff})$	S-35	S-35, S-139, S-142	n/a
4.5	SPARC	$S_\theta(\omega)=(\hbar/e)^2 F S_I(\omega)/(\gamma_{rot}^2+I^2\omega^2)$	S-15	S-15, S-3, S-128	n/a
5.1	TURTLE	$\tau_{load}=\tau_{drive}-k_BT\sigma/\Omega$	S-38	S-38, S-132, S-12	Agreement with optical trap load within TUR bound
5.2	TURTLE	$\varepsilon_\tau^2 \text{equiv } Var(\tau)/\langle\tau\rangle^2 \geq 2k_B/\sigma\, t_{win}$	S-52	S-52, S-38, S-137	n/a
5.3	TURTLE	$t_{win}^{min}=2k_B/\sigma\varepsilon^2$	S-52	S-52, S-38	n/a
5.4	TURTLE	$\ln P(+\Delta\theta)/P(-\Delta\theta)=(\tau_{drive}-\tau_{load})\Delta\theta/k_BT$	S-94	S-94, S-38, S-52	n/a
5.5	TURTLE	$Alarm=1[\sigma < 2k_B\, Var(J)/J^2\, t_{win}]$	S-52	S-52, S-94, S-37	n/a
6.1	GYRE	$\gamma_B(m)=\oint Im\langle m(\theta)\|\partial_\theta\, m(\theta)\rangle\, d\theta=2\pi\, m\, \chi_g$	S-131	S-131, S-57, S-58	n/a
6.2	GYRE	$\varepsilon_m=\alpha_{SO}\, m\langle S_z\rangle$	S-67	S-67, S-57, S-58	n/a
6.3	GYRE	$A(t)=\Sigma_m\, p_m\, e^{-iE_m\, t/\hbar}$, $E_m=\hbar^2\, m^2/2I+\varepsilon_m$	S-68	S-58, S-67, S-68	n/a
6.4	GYRE	$delta\gamma_B/\gamma_B = O((delta\ t/T_{cyc})^2)$	S-131	S-131, S-57, S-68	Phase invariant under write-cycle time deformation
6.5	GYRE	$t_{ret}\approx Gamma_phi^{-1}$, $Gamma_phi\propto\int_0^\infty d\omega\ \omega\ J(\omega)\ \coth(\hbar\omega/2k_BT)$	S-69	S-69, S-58, S-131	n/a
7.1	TEMPO	$k_{dose}(V)=k_0\exp(-(\Delta U_0-q_{eff}V)/k_BT)$	S-35	S-35, S-88, S-130	Arrhenius slope shifts with V as (7.1)
7.2	TEMPO	$P(t)=k_{dose}\, e^{\wedge}-k_{dose}t$, $CV_N=1/\sqrt{(N)}$	S-88	S-88, S-86, S-35	n/a
7.3	TEMPO	$\langle C\rangle_{ss}=ND_0k_{dose}/V_d\, k_{el}$, $\sigma_C/\langle C\rangle\approx sqrtk_{el}/2Nk_{dose}$	S-88	S-88, S-86, S-37	n/a
7.4	TEMPO	$k_{dose}\approx k_{cl}\kappa_{tun}(T)$, $\kappa_{tun}\to 1\ (T\ll T_{cross})$	S-90	S-90, S-35, S-86	n/a
7.5	TEMPO	$Dose_{permit}=1[\sigma_{local}\leq\sigma_{tissue}]$	S-37	S-37, S-132, S-130	n/a
8.1	CHILL	$e\|V\|<\hbar\Omega_{res}$	S-7	S-7, S-6, S-104	Teff below bath across threshold
8.2	CHILL	$Gamma_{cool}=Gamma_{IET}(-\Omega_{res},V)$, $Gamma_{heat}=Gamma_{IET}(+\Omega_{res},V)$	S-6	S-6, S-7, S-130	n/a
8.3	CHILL	$n_f=(\gamma\, n_B+Gamma_{heat})/(Gamma_{cool}-Gamma_{heat})$, $n_B=1/(e^{\wedge}\hbar\Omega_{res}/k_BT-1)$	S-130	S-105, S-106, S-107, S-130	n/a
8.4	CHILL	$COP\leq\hbar\Omega_{res}/e\|V\|$	S-135	S-135, S-127, S-6	n/a
8.5	CHILL	$T_{eff}(\omega_{res})=S_\tau(\omega_{res})/2k_B\, Re\gamma_{rot}(\omega_{res})$	S-142	S-142, S-104, S-130	FDT-violation thermometry
9.1	ONSET	$I=L_{11}V+L_{12}\tau+L_{13}F$, $\Omega=L_{21}V+L_{22}\tau+L_{23}F$, $U=L_{31}V+L_{32}\tau+L_{33}F$	S-134	S-134, S-79, S-123	n/a
9.2	ONSET	$L_{ij}=L_{ji}\Rightarrow I/F=L_{13}=L_{31}=U/V$	S-80	S-80, S-77, S-134	n/a
9.3	ONSET	$Z=L_{13}^2/L_{11}L_{33}$, $\eta_{max}=(1-\sqrt{(1-Z^2)})/(1+\sqrt{(1-Z^2)})$	S-134	S-43, S-135, S-134	n/a
9.4	ONSET	$eV_{out}I_{out}\leq F_{flow}U_{flow}-\gamma_{rot}\Omega^2-Qdot$	S-127	S-127, S-123, S-43	n/a
9.5	ONSET	$FoM=L_{13}^2/(L_{11}L_{33}-L_{13}^2)$	S-134	S-134, S-52, S-43	Cross-coefficients equal on same device
10.1	TORCH	$b=sgn(\langle L_z\rangle)$, $b\leq ftarrow\ sgn(V-V_0)$	S-10	S-10, S-9, S-21	n/a
10.2	TORCH	$Q_{flip}\geq k_BT\ln p_{CW}/p_{CCW}$	S-41	S-41, S-10, S-35	n/a
10.3	TORCH	$t_{ret}=2\pi\gamma_{rot}/\omega_0\omega_b\exp(\Delta U/k_BT)$	S-35	S-35, S-21, S-23	n/a
10.4	TORCH	$b_{out}=sgn(\Sigma_{i=1}^3\, w_i\, b_i)$	S-123	S-123, S-21, S-9	n/a
10.5	TORCH	$\tau_{out}/\tau_{in}=(R_{out}/R_{in})^3$, $\gamma=8\pi\eta_f R^3$	S-23	S-23, S-123, S-21	3-rotor majority truth table 99%
11.1	ANGEL	$Gamma_{ent}(m\to m')=Gamma_{IET}\, FC_{m\to m'}$, $FC=\|\langle m'\|e^{ip_\theta\Delta\theta/\hbar}\|m\rangle\|^2$	S-71	S-65, S-71, S-70	n/a
11.2	ANGEL	$Flag=1[\Delta L_{el}+\Delta L_{rot}\neq 0]$	S-66	S-66, S-65, S-71	Conservation syndrome, no ancilla
11.3	ANGEL	$dS_{ent}/dt=Gamma_{ent}\, bar\, E_{FC}-Gamma_phi\, S_{ent}$, $S_{ent}^{ss}=Gamma_{ent}\, bar\, E_{FC}/Gamma_phi$	S-69	S-69, S-70, S-138	n/a

ID	Tech	Formula	Load-bearing	Ancestors	Falsifier / claim
11.4	ANGEL	$C=\max_V[H(m)-H(m\|m')]\leq\log_2(2m_{max}+1)$	S-138	S-138, S-71, S-69	n/a
11.5	ANGEL	$F_{ro}=(1+Gamma_phi/2Gamma_{ent})^{-1}$	S-69	S-69, S-70, S-138	Positive entanglement by tomography
12.1	QUANTA	$\tau_{std}=\hbar/e\ \eta(V)I,\ \eta(V^*)\to1$	S-12	S-12, S-2, S-57	n/a
12.2	QUANTA	$\langle L_z)(V)=m\hbar$ on $\Delta V_p,\ \Delta V_p>>u_{noise}$	S-57	S-2, S-57, S-11	n/a
12.3	QUANTA	$\tau_{std}=\hbar/e\ I_{JQHR}=n\ h\ f/4\pi$	S-12	S-12, S-18, S-57, S-135	tau = n h f / 4pi traceable to SI second
12.4	QUANTA	$f_{DUT}=\tau_{std}/\Omega_{DUT}$, $u_r(f_{DUT})=sqrtu_r^2(\tau_{std})+u_r^2(\Omega_{DUT})$	S-12	S-2, S-12, S-52	n/a
12.5	QUANTA	certify(η=1)Leftrightarrow $Var(L_z)/\hbar^2\to0$	S-52	S-52, S-137, S-2	Plateau flatness 1e-3 at 4K; material independence
13.1	AEGIS	F_{soma}(π)=($E_{harm}(\pi)+\lambda\ C_{comp}(\pi)$)/Efficacy(π)	G-133	G-133, G-61, S-37	n/a
13.2	AEGIS	Adopt=1[ΔF_{soma}<0 ∧ ∀ i: $\Delta\ g_i\geq0$]	G-61	G-61, G-66, G-71	n/a
13.3	AEGIS	g=(pH,T,G_{glu},C_{cyt},v_{flow}), ∀ i: $g_i\geq g_i^{min}$	G-66	G-66, S-37, S-132	n/a
13.4	AEGIS	$\|\sigma_{meas}-\sigma_{log}\|\leq\varepsilon_{th}$	S-132	S-37, S-39, S-132, G-1	Heat audit catches log fraud (red-team drill)
13.5	AEGIS	$t_{recall}=t_{dec}+d/v_{retro}\leq\tau_{max}$	G-71	G-71, G-24, S-124	Red-team rollback 99.9%
14.1	CODEX	A=$M_1..M_{12}$,$P_1..P_3$, $\rho_{mol}=H(X_{prose})/H(X_{code})=H_{prose}/\log_2$ 15	G-111	G-111, G-114, G-119	n/a
14.2	CODEX	$Var(t_{arr})\approx 2D_{trans}d/U^3$	S-125	S-116, S-125, S-136	n/a
14.3	CODEX	$R=\lambda_s\ \log_2(15)[1-erfc(t_{sym}/2\sqrt{(2)}\sigma_t)]$	S-125	G-106, G-114, S-125	Error-rate curve matches (14.3) with jitter from (14.2) only
14.4	CODEX	$P_{undetected}\leq(1-V_{schema})^2$	G-114	G-114, G-115, G-116	n/a
14.5	CODEX	$E_{claim}\geq k_BTln2\ H_{bits}/\rho_{mol}$	G-114	G-41, G-114, S-37	Landauer bound / codebook gain

Table A1 — The 60 base equations [CANDIDATE].

Appendix B Enhancements (48) and Drill Ledger

The 48 enhancements from *07-NOVEL-ENHANCEMENTS-48.md*, with their readiness tags (NEAR = months / Forge simulator, MID = junction or clinical program, HORIZON = gated on CHILL ground-state prep or in-body operation) and cheapest-kill lines, followed by the measured red-team ledger.

ID	Name	Tag	Cheapest kill
3.6	TWIN	NEAR	Heading linear in differential current; reverses at V0
3.7	SHIFT	MID	Two-slope F-I curve with hysteresis at pitch-switch voltage
3.8	GOV	NEAR	Step-load stall exactly at Omega_SO/2, no runaway
3.9	REGEN	HORIZON	Negative current signature under enforced flow reversal
4.6	LITHOS	HORIZON	IETS m-distribution matches engineered noise spectrum, zero intercept
4.7	THERM	NEAR	Agreement with independent fluorescence thermometry within bound
4.8	KEY	MID	Fingerprint repeatability over 10^4 cycles; drift > threshold kills
4.9	STOCH	HORIZON	Delta-SNR curve peaks at predicted noise tone; no peak kills
5.6	VISC	NEAR	Glycerol-gradient staircase tracked within 5.2 bound

ID	Name	Tag	Cheapest kill
5.7	SONAR	NEAR	Titration shows quantized d tau_load steps at known Kd
5.8	LEDGER	MID	Map integral matches phantom microcalorimetry within eps_th
5.9	COMPASS	MID	Deliberate flow reversal flips sign as predicted
6.6	SHIELD	HORIZON	Induced staple defect does not flip winding; Arrhenius in T_mode
6.7	MULTI	MID	All m <= m_max read with fidelity > 1-10^-3
6.8	GYRO	MID	Delta gamma_B linear in Omega_frame, zero intercept, flips with handedness
6.9	WITNESS	HORIZON	n_f unchanged after readout within delta_nd; else demolition kills
7.6	TITRAT	NEAR	Glucose step settles within bound, no guard-floor breach
7.7	GRAD	MID	Fluorescent gradient velocity matches v_g prediction
7.8	PERMIT	NEAR	Refusals begin exactly at predicted sigma threshold under heat load
7.9	FLOWZERO	MID	Release constant to <5% over decade of flow
8.6	CASCADE	HORIZON	Two-sideband suppression ratio matches n_f2 proportional n_f1
8.7	DUTY	MID	n_f vs duty curve minimum at predicted D*
8.8	FINGER	HORIZON	Drot vs T_eff linear with FDT slope
8.9	STEALTH	HORIZON	Phantom calorimetry shows predicted tissue-side delta sigma
9.6	HARVEST	MID	V_oc proportional v_flow^2, exponent 2.0+-0.1 in mock loop
9.7	SENTINEL	NEAR	Aged junctions flagged before thrust failure in 10^3-cycle test
9.8	Z2ANG	HORIZON	Measured eta_max matches transplanted formula across loads
9.9	SHARE	HORIZON	Round-trip efficiency decays as exp(-d/lambda_ion)
10.6	WEIGHT	NEAR	Weighted truth table >=99% over input set
10.7	LATCH	NEAR	Arrhenius t_ret vs 1/T slope yields designed dU
10.8	REV	HORIZON	E_op linear in 1/T_cyc with Landauer intercept
10.9	HARD	MID	BER ratio <10^-3 vs charge memory at same gamma dose
11.6	SYNDROME	HORIZON	Injected +-hbar errors corrected >=99%, no ancilla
11.7	ROUTE	MID	Propensity shifts scale as sqrt(M1/M2) mass weighting
11.8	PARK	HORIZON	Retention ratio at clock point matches curvature model
11.9	TICK	MID	Cross-correlation width meets 14.2 jitter bound
12.6	BALANCE	NEAR	Force agrees with conventional chain within combined u_r
12.7	COMB	MID	Comb spacing matches independent mass spectrometry
12.8	KELVIN	HORIZON	4 K He-scale and triple-point checks agree
12.9	SENATE	NEAR	Injected drifted unit detected 99.9% in one window
13.6	WINDOW	NEAR	ROC AUC on fraud ladder matches d-prime model
13.7	TWINP	NEAR	Invalid twin rejected; floors frozen; valid twin predicts guard trajectory
13.8	DRILL	NEAR	Any drill exceeding tau_max fails the certificate
13.9	PRIV	HORIZON	99% audit power retained at eps_p = 1 on synthetic cohort
14.6	BINS	NEAR	R(B) peaks at predicted B*; erfc model fits
14.7	TITER	MID	Dose-response curves separable across rho_R tiers
14.8	ARQ	MID	Reliability-vs-energy curve matches E[N] = 1/P_pass
14.9	EVOLVE	HORIZON	Transition-window error spike <= 2x steady state

Table B1 — The 48 enhancements [CANDIDATE] (16 NEAR / 16 MID / 16 HORIZON).

Drill	Covers	Injection	Outcome	Measured
LOG_FRAUD_SINGLE	13.4/13.6	unit 7 shaves 35% of its sigma (share 2.6%)	caught	effect 0.009 of fleet total; ~24 windows to 3-sigma — honestly priced, not hidden (13.6)
LOG_FRAUD_COORDINATED	13.4	6 units shave 35% each (share 23%)	caught	39/40 windows over budget — detected in-window
ROGUE_POLICY	13.2/13.3	policy improves F_soma by overheating tissue (guard breach)	caught	adoption=False (must be False); reason: rejected: dF=+0.0046, guard breach at pH
SPOOFED_TELEMETRY	13.3	unit reports pH 7.42 while boundary sensing reads 6.72	caught	guard cross-check flags divergence beyond tolerance
JAMMED_LINK	14.4	burst jamming lifts per-symbol error to 0.90	caught	parity rejected 1865/2000 corrupted grams; undetected 135 (bound 14.4)
RECALL_STRESS	13.5/13.8	1000 rollback drills vs tau_max = 12.0	not caught	CERTIFICATE NOT ISSUED: 12/1000 drills exceeded tau_max, worst 14.9; certifying tau_max >= 14.9 (13.8: no statistical rescue)

Table B2 — Red-team drill ledger with G-65 coverage ids (seed 20261102).